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Some Notes on the Role of Field Representations in Electromagnetic Information Theory

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***Abstract*—Electromagnetic Signal and Information Theory (ESIT) is an emerging field that integrates electromagnetic and information theories to explore the constraints imposed by the laws of electromagnetism on wireless telecommunications systems.**

This contribution will discuss four different representations of the field in free space. From an electromagnetic point of view, they are equivalent, all allowing an accurate representation of the field. From the ESIT perspective, they play quite different roles, focusing in some cases on field information and other cases on the physical implementation of the ESIT.

I. MODEL AND FIELD REPRESENTATIONS

Let us consider a simple 2D electromagnetic model consisting of harmonic electromagnetic sources in the D domain placed in the half-plane $z \leq 0$ and observed in a domain Ω placed in $z > 0$ half-space (Fig. 1). The field is directed along $\hat{y}$ and $\partial/\partial y = 0$. The boundary $z = 0$ can be a real or virtual surface, f.i. placed at a small distance from a periodic structure.

A. Hilbert-Schmidt representation

The field $E(x, z)$ is related to the current density distribution $J(x, z)$ via the radiation integral [1]:

$$E(x, z) = \int_D G(x, z; x', z')J(x', z')dx'dz' \qquad (1)$$

wherein G is the Green's function, and E and J are square integrable functions.

Due to the compactness of the integral operator, it can be diagonalized by Hilbert-Schmidt decomposition obtaining:

$$E(x, z) = \sum_{k=0}^{NDF_\epsilon} \sigma_k < J(x', z'), v_k(x', z') > u_k(x, z) \qquad (2)$$

wherein (σ_k, u_k, v_k) is the singular system of the integral operator, $\sigma_1 \geq \sigma_2 \geq \sigma_n \geq ... > 0$ are the singular values of the operator, and NDF_ϵ, called the Number of Degrees of Freedom of the field at approximation level ϵ, is the minimum number of basis functions to obtain an approximation error lower than ϵ [2].

The Hilbert-Schmidt representation is the main tool of the ESIT. In particular, NDF_ϵ represents a "bridge" between electromagnetic theory and information theory. In fact, on the one hand, it reflects in a very compact way some fundamental constraints imposed by Maxwell's equation on any radiant source; on the other hand, it limits the amount of information that wireless communication systems can reliably transmit by setting an upper bound on the number of spatial subchannels in a MIMO [3] system.

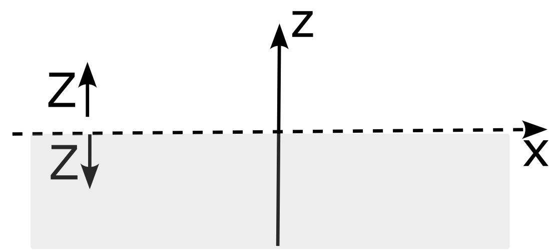

Fig. 1. Geometry of the problem. The source is placed in the $z \leq 0$; the field is observed in $z > 0$; $Z\downarrow(\omega))$ and$Z\uparrow(\omega)$ are respectively the modal impedance associated to the "lower" region $z < 0$ containing the sources and the free-space modal impedance, evaluated at the interface $z = 0$.

B. Minimum redundant bandlimited representation (MRBR)

The Shannon-Whitaker sampling series is one of the most straightforward and practically helpful field representations. It is understood that the sampling representation is not generally optimal since, for a given approximation error, the number of samples is larger than the NDF_ϵ. So, obtaining a sampling representation of the field as close as possible to the best one is of great interest.

This problem can be solved by introducing a transformation of the field on the observation domain, which, the for sake of simplicity, will be supposed to be a curve [4]:

$$F(\xi(x, z)) = E(\xi(x, z))e^{j\psi(\xi(x,z))} \qquad (x, z) \in \Omega \qquad (3)$$

wherein $\xi(x, z)$ is a parameterization function along the observation curve, and $\psi(\xi(x, z)))$ is a proper phase function.

The function $F(\xi(x, z))$ is approximated by the function $F_w(\xi(x, z))$ bandlimited to w. The theory identifies the ξ and ψ functions allowing to minimize the bandwidth w keeping the bandlimitation approximation error within a desired value [4].

The number of samples turns out to be only slightly larger than NDF_ϵ. In particular, the curves passing through the points $\xi_n = n\pi/w$, wherein $n \simeq NDF_\epsilon$, are associated with independent flows of spatial information and allow us to have a graphical representation of the "flows of spatial information" in the space around the antenna [3].

C. Plane-wave representation

The classic plane-wave representation of the field in the half-space $z > 0$, referred as "longitudinal spectral representation" in [5], is obtained by integrating along the wavenumber k_x [5]:

$$E(x,z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(k_x) \exp[-j(k_x x + k_z z)] dk_x \tag{4}$$

wherein

$$f(k_x) = \int_{-\infty}^{\infty} E(x, z=0) \exp[jk_x x] dx \tag{5}$$

and $k_z = \sqrt{\beta^2 - k_x^2}$ if $k_x^2 \leq \beta^2$, $k_z = -j\sqrt{k_x^2 - \beta^2}$ if $k_x^2 > \beta^2$, $\beta = 2\pi/\lambda$, λ being the free space wavenumber.

The integral represented by Eq. 4 is a continuous superposition of plane waves and exhibits the characteristics of a Fourier transform. The region where $k_x^2 \leq \beta^2$ is associated with waves carrying active power, and is referred to as the "visible domain". Beyond this range, the term $\exp[-jk_z z]$ exhibits exponential decay by moving away from the plane $z = 0$. This portion of the spectrum is associated with evanescent waves and is called the "invisible domain". The spectral components within the invisible domain do not contribute to the far-field, but they hold a crucial role in the close vicinity of the aperture, influencing the reactive energy surrounding the source.

D. Other representations

In the "transverse" spectral representation ([5]) Eq. 4 is transformed to obtain a representation of the field in terms of superposition of $e^{-jk_z z}$ contributions instead of $e^{-jk_x x}$ and modifying the integration path on the k_z complex plane.

The new integration path suitably takes into account the branch points in $k_z = \pm\beta$, by properly defining the branch-cuts, and the proper poles within the contour integral, which are given by the solutions of the transverse resonance equation [5]:

$$Z\downarrow(k_z) + Z\uparrow(k_z) = 0 \tag{6}$$

wherein $Z\downarrow(k_z)$ and $Z\uparrow(k_z)$ are respectively the modal impedance associated to the "lower" region $z < 0$ containing the sources and the free-space modal impedance, evaluated at the interface $z = 0$ (Fig. 1). The transverse spectral representation is particularly useful to classifying the spectrum of guided waves, given by poles associated with real solutions of Eq. 6. Consequently, loosely speaking the field $E(x,z)$ is represented by a continuous spectrum arising from the cuts, in addition to a discrete set of waves associated with poles that lie within the contour of integration.

Different deformations of the integration path give different field representations that highlight different field characteristics. Among them, the Steepest-Descent (SD) representation allows to obtain an effective asymptotic expressions of the far-field radiated by the structure. The SD representation also highlights complex solutions of Eq. 6 (leaky-waves [5]) that may contribute to the field representation based on the observation angle.

II. CONCLUSIONS

Different representations of the electromagnetic field highlight different characteristics of the field.

Representations based on Hilbert-Schmidt decomposition, MRBR and plane wave expansion directly link to the amount of information transmittable by the radiating structure. While Hilbert-Schmidt decomposition gives an optimal representation, MRBR simplifies the analysis at the cost of a slight increase in the number of basis functions. MRBR allows the direct application of numerous results obtained for band-limited functions in information theory, and provides a simple graphical representation of the spatial flows of information around the source [3]. Thanks to the Fourier transform relationship between currents and field at large distance from the source, the plane wave expansion allows an easy analysis of the far-field information content. It is worth noting that other representations of the field, as spherical mode expansion and characteristic modes, are of interest and worth further investigation in the ESIT.

Further types of representations are crucial in analyzing the set of field configurations that specific radiating structures can generate. These representations are of particular importance when studying structures such as smart reflective surfaces [6] or spatial information lenses [7]. However, the links between these representations and the information content of the electromagnetic field have received limited investigation and remain ambiguous. Clarifying these relationships would be highly beneficial, as it would provide new tools in ESIT that are closely aligned with actual electromagnetic systems.

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